

DISTRIBUTIONAL DYNAMICS

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MOTIVATION

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Commonly used U.S. microdata — PSID, CEX, SCF, CPS, SIPP — each covers only part of the picture.

WHAT WE DO

We develop a method that produces **consensus estimates** of joint distributions of consumption, income, and wealth ...

- ▶ ... at a **balanced, high frequency** (quarterly);
- ▶ ... for **any granularity** (deciles, joint cells);
- ▶ ... over a **long horizon** (1962–2024);
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We validate the method three ways and apply it to consumption dynamics across the last three U.S. recessions.

WHERE WE SIT

We speak to four strands:

- ▶ **HA macro:** [kaplan2018monetary; Bayeretal2019; bayer2020shocks]
- ▶ **Inequality time-series:** [piketty2018distributional; batty2020distributional; blanchet2022real]
- ▶ **Temporal disaggregation:** [harvey2000estimating; proietti2006temporal]
- ▶ **Functional data in economics:** [kneip2001inference; chang2024heterogeneity]

Closest paper: **Chang, Chen, and Schorfheide (JPE, 2024)** — earnings marginal dynamics. We extend to **joint** C , Y , W and emphasize handling missing/mixed-frequency microdata.

FUNCTIONAL DYNAMICS, THREE DIFFICULTIES

Let $\Xi_t(w)$ be the joint distribution over $w = (c, y, w)$ and Y_t aggregate variables:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

Three difficulties to overcome:

- (1) Density $d\Xi_t$ has **time-varying support**;
- (2) w is **multi-dimensional**, observed *not* $\forall t$, and at times **incompletely**;
- (3) The **high-dimensional functional** difference equation must be feasible.

(1) DECOMPOSE INTO COPULA + QUANTILE FUNCTIONS

By Sklar's theorem, the joint splits into:

$$dC_t(u_1, \dots, u_d) \quad \text{and} \quad \Xi_{mt}^{-1}(u_m), \quad u_m \sim U[0, 1]$$

- ▶ dC_t : copula density — captures the **dependence structure**;
- ▶ Ξ_{mt}^{-1} : quantile functions — capture **marginal shapes**.

Why this helps: both objects live on the unit hypercube $\Rightarrow$ **time-invariant support**.

LEGENDRE POLYNOMIAL EXPANSION

$$dC_t(u_1, \dots, u_d) = \sum_{o \in \mathbb{N}^d} \kappa_{o,t} \prod_{m=1}^d Q_{o_m}(u_m)$$

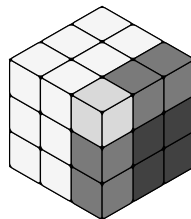
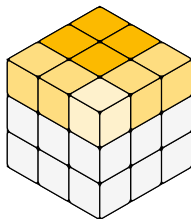
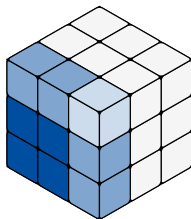
$$\Xi_{mt}^{-1}(u_m) = \sum_{o \in \mathbb{N}} \xi_{o,m,t} Q_o(u_m)$$

By uniformity and orthonormality:

$$\kappa_{o,t} \approx N^{-1} \sum_i \prod_{m=1}^d Q_{o_m}(u_{m,i,t}), \quad \xi_{o,m,t} \approx N^{-1} \sum_i w_{m,i,t} Q_o(u_{m,i,t})$$

Truncate to a finite set $\mathbb{O} \Rightarrow$ tracking the distribution = tracking finitely many coefficients $\hat{\phi}$.

(2) PARTIAL OBSERVABILITY HANDLES ITSELF



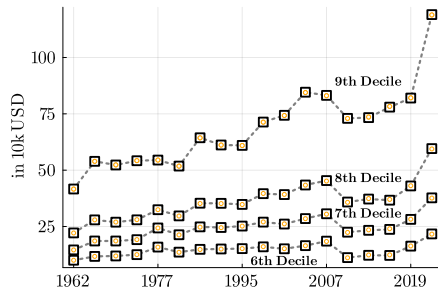
- ▶ A dataset missing one dimension still informs the *lower-dim slice* of the copula;
- ▶ Leading-order coefficients of the d -dim copula coincide with the $(d-1)$ -dim copula.

(3) DIMENSIONALITY REDUCTION VIA FACTOR MODEL

The coefficient vector $\hat{\phi} = \{\xi, \kappa\}$ is still high-dimensional. Decompose via SVD on standardized $\hat{\phi}$:

$$\hat{\phi} \approx \Gamma F$$

- ▶ Γ : projection onto an ideal lower-dimensional subspace;
- ▶ F : a small number of latent factors.



Wealth quantiles: raw vs. factor reconstruction $\Rightarrow$ 8 factors explain $\sim 99\%$

LINEAR-GAUSSIAN STATE-SPACE MODEL

$$\begin{aligned}
 F_{t+1} &= AF_t + BY_{t+1} + \epsilon_{t+1}, & \epsilon_{t+1} &\sim \mathcal{N}(0, \Omega) \\
 \hat{\phi}_{t+1}^j &= H_{t+1}^j (\Gamma F_{t+1} + v_{t+1}^j), & v_{t+1}^j &\sim \mathcal{N}(0, \Sigma_j^{1/2} \Delta_j \Sigma_j^{1/2'})
 \end{aligned}$$

- ▶ F_t : small set of latent factors driving the distribution;
- ▶ Y_t : aggregate variables (FRED-QD, balance sheets, prices ...);
- ▶ H_{t+1}^j : selector — handles **mixed frequency** and **partial observability**;
- ▶ Σ_j : sampling VCV $\Rightarrow$ measurement noise anchored to data uncertainty.

ESTIMATION: BAYESIAN + KALMAN SMOOTHER

Priors (Minnesota-type on A, B ; log-normal on variances):

- ▶ shrinkage handles the dense parameter space and data sparsity;
- ▶ posterior draws give credible bounds on the synthetic data.

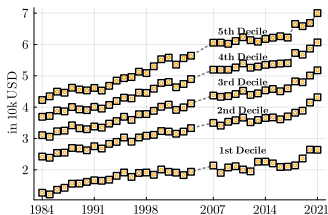
Reconstruction via the Kalman smoother:

$$\hat{F}_t \xrightarrow{\Gamma} \tilde{\phi}_t \xrightarrow{\text{un-standardize, add trend}} \hat{\phi}_t \rightarrow \hat{\Xi}_t(w)$$

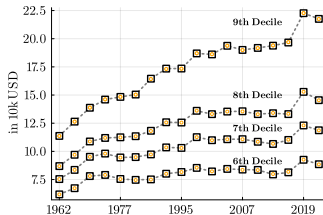
$\Rightarrow$ **synthetic microdata** at quarterly frequency for any t — even where no micro waves exist.

VALIDATION 1: FACTOR APPROXIMATION IS NEAR-LOSSLESS

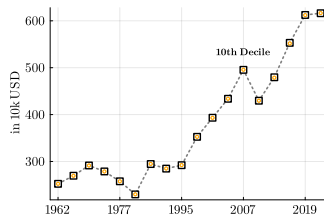
Quantile functions: raw data vs. 8-factor approximation



Consumption (bot.)



Income (mid.)

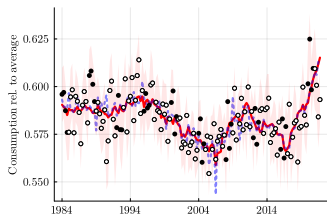


Wealth (top)

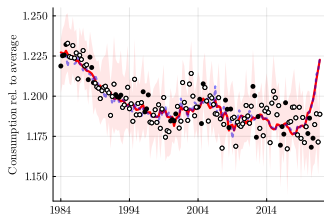
- ▶ 8 factors explain $\approx 99\%$ of business-cycle variation in $\hat{\phi}$;
- ▶ KL-divergence of approximated vs. raw copula is $\sim 100\times$ smaller than the divergence captured by time variation.

VALIDATION 2: PREDICTING OMITTED SURVEY WAVES

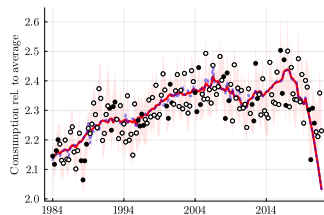
CEX every 4 years (mimicking European frequency)



Bottom 50



Next 40



Top 10

- ▶ Black dots: observations *kept*; empty dots: *dropped* from the smoother;
- ▶ Solid line: baseline (all data); dashed: smoother with only 1/4 of CEX waves;
- ▶ Aggregates + surrounding microdata fill the gaps.

VALIDATION 3: HANK MONTE CARLO

Use **Bayer–Born–Luetticke (2024)** HANK as DGP → draw 4 survey-style samples (PSID/CEX/SCF/CPS-like) → compare DDFM estimates to the simulated truth.

Time-series correlation with truth, averaged across 10 economies

Group	<i>Consumption</i>		<i>Income</i>		<i>Wealth</i>	
	DDFM	Linear	DDFM	Linear	DDFM	Linear
Bottom 50	96	84	98	72	96	96
Middle 50–90	97	84	94	49	81	79
Top 10	93	42	81	45	87	31

- ▶ DDFM beats naive linear interpolation *everywhere*;
- ▶ Gains largest where data is scarcest (top tail, low-frequency wealth).

APPLICATION 1: AGGREGATE SHOCKS DOMINATE

Forecast-error variance decomposition on the 8 distributional factors (5-yr horizon, blocked Cholesky, aggregates first):

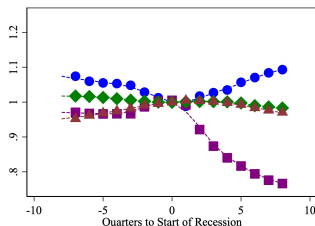
Factor No.	1	2	3	4	5	6	7	8
Aggregate shocks	99	99	70	89	93	86	86	77
Distributional shocks	1	1	30	11	7	14	14	23

- ▶ Aggregate shocks explain $\geq 70\%$ of every factor (typically $>85\%$);
- ▶ Same pattern holds for conditional means of C , Y , W across deciles;
- ▶ Echoes [bayer2020shocks]: aggregates shape distributions.

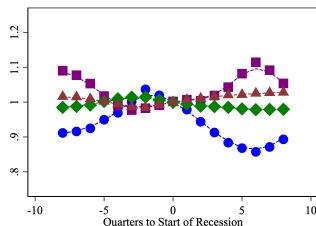
APPLICATION 2: CONSUMPTION IN THREE RECESSIONS

Relative consumption indexed to recession peak ($t = 0$), by **joint income** $\times$ **wealth** groups

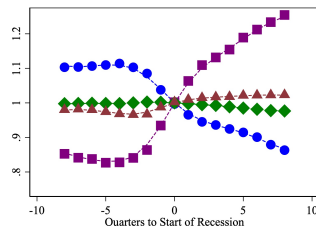
Dot-com bust



Global Financial Crisis



COVID-19



4 groups: **liq.-constrained** (low Y, low W) • **asset-rich low-Y** (low Y, high W) • **middle**
• **rich** (high Y, high W)

► Distributional consequences differ **sharply** across recessions.

TAKE-AWAY: STATE-DEPENDENCE IN RECESSIONS

- ▶ **Dot-com:** a *wealth* shock $\Rightarrow$ asset-rich low- Y households hit hardest;
- ▶ **Global Financial Crisis:** a *balance-sheet* shock $\Rightarrow$ low-wealth, indebted households suffer the largest declines;
- ▶ **COVID-19:** an *income + transfers* shock $\Rightarrow$ bottom 20% *gains* $\sim$ 10% relative to average.

The joint distribution matters — the relevant channel is recession-specific.
Single-margin elasticities mask this state-dependence.

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- ▶ Recession-time consumption losses are **state-dependent** along the joint $Y \times W$ distribution;

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- ▶ **Aggregate shocks** drive the bulk of distributional fluctuations;
- ▶ Recession-time consumption losses are **state-dependent** along the joint $Y \times W$ distribution;
- ▶ New empirical targets for HANK models; real-time diagnostic for distributional policy.

(2) DEALING WITH PARTIAL OBSERVABILITY OF w

To estimate all $\kappa_{o,t}$, we need to observe all variables $m \in \mathbb{D}$ in dataset j at time t .

What if we only observe $\mathcal{D}^j \subset \mathbb{D}$? Can we still inform the estimation of κ_t^j ?

$$dC(u_1, \dots, u_{d-1}) = \int_0^1 dC(u_1, \dots, u_d) du_d = \int_0^1 \sum_{o \in \mathbb{O}} \kappa_o \prod_{m=1}^d Q_{o_m}(u_m) du_d$$

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because $\int_0^1 Q_{o_d}(u_d) = \begin{cases} 1 & \text{for } o_d = 1, \\ 0 & \text{otherwise} \end{cases}$

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→ the leading coefficients in dimension d and the coefficients from the $(d-1)$ -dim copula are identical.